

Practical 2

Write a Python program to implement Priority CPU Scheduling and calculate Waiting Time and Turnaround Time.

Code :

```
def get_positive(msg):
    while True:
        try:
            x = int(input(msg))
            if x > 0:
                return x
            print("Invalid input! Enter a positive number.")
        except ValueError:
            print("Invalid input! Enter numbers only.")

n = get_positive("Enter number of processes: ")
bt = []
priority = []
process = []
for i in range(n):
    process.append(i+1)
    bt.append(get_positive("Enter Burst Time for P" + str(i+1) + ": "))
    priority.append(get_positive("Enter Priority for P" + str(i+1) + ": "))
for i in range(n):
    for j in range(i+1, n):
        if priority[i] > priority[j]:
            priority[i], priority[j] = priority[j], priority[i]
            bt[i], bt[j] = bt[j], bt[i]
            process[i], process[j] = process[j], process[i]

wt = [0]
total_wt = 0
total_tat = 0
print("\nProcess\tPriority\tBT\tWT\tTAT")
for i in range(1, n):
    wt.append(wt[i-1] + bt[i-1])
for i in range(n):
    tat = wt[i] + bt[i]
    print("P" + str(process[i]), "\t", priority[i], "\t\t", bt[i], "\t",
wt[i], "\t", tat)
    total_wt += wt[i]
    total_tat += tat
print("\nAverage Waiting Time =", total_wt/n)
```

```
print("Average Turnaround Time =", total_tat/n)
```

Output :

```
Enter number of processes: 3
Enter Burst Time for P1: 3
Enter Priority for P1: 5
Enter Burst Time for P2: 6
Enter Priority for P2: 4
Enter Burst Time for P3: 8
Enter Priority for P3: 1
```

Process	Priority	BT	WT	TAT
P3	1	8	0	8
P2	4	6	8	14
P1	5	3	14	17

```
Average Waiting Time = 7.333333333333333
Average Turnaround Time = 13.0
```